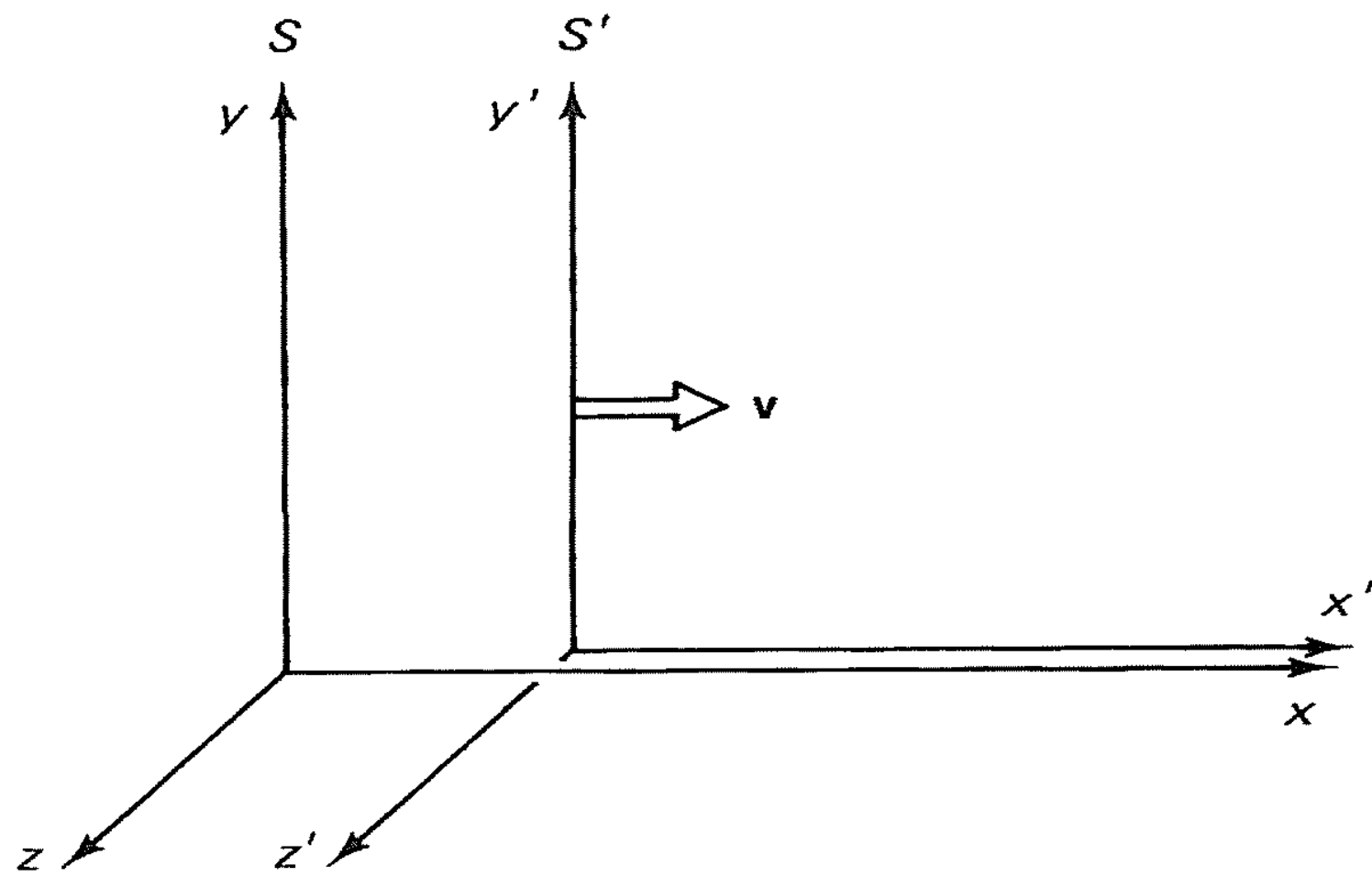


Introduction to Particle Physics

Special Theory of Relativity

Lorentz Transformations

$$\gamma \equiv \frac{1}{\sqrt{1 - v^2/c^2}}$$



Velocity addition: Suppose a particle is moving in the x direction at speed u' , with respect to S' . What is its speed, u , with respect to S ? Well, it travels a distance $\Delta x = \gamma(\Delta x' + v \Delta t')$ in a time $\Delta t = \gamma[\Delta t' + (v/c^2)\Delta x']$, so

$$\frac{\Delta x}{\Delta t} = \frac{\Delta x' + v \Delta t'}{\Delta t' + (v/c^2)\Delta x'} = \frac{(\Delta x'/\Delta t') + v}{1 + (v/c^2)(\Delta x'/\Delta t')}.$$

But $\Delta x/\Delta t = u$, and $\Delta x'/\Delta t' = u'$, so

$$u = \frac{u' + v}{1 + (u'v/c^2)}$$

$$\text{i. } x' = \gamma(x - vt)$$

$$\text{ii. } \gamma' = \gamma$$

$$\text{iii. } z' = z$$

$$\text{iv. } t' = \gamma \left(t - \frac{v}{c^2}x \right)$$

$$\text{i'. } x = \gamma(x' + vt')$$

$$\text{ii'. } \gamma = \gamma'$$

$$\text{iii'. } z = z'$$

$$\text{iv'. } t = \gamma \left(t' + \frac{v}{c^2}x' \right)$$

$$u_x = \frac{x}{t} = \frac{u'_x + v}{1 + vu'_x/c^2}, \quad u_y = \frac{y}{t} = \frac{u'_y \sqrt{1 - v^2/c^2}}{1 + vu'_x/c^2},$$

$$u_z = \frac{z}{t} = \frac{u'_z \sqrt{1 - v^2/c^2}}{1 + vu'_x/c^2}.$$

$$u'_x = \frac{x}{t} = \frac{u_x - v}{1 - vu_x/c^2}, \quad u'_y = \frac{y}{t} = \frac{u_y \sqrt{1 - v^2/c^2}}{1 - vu_x/c^2},$$

$$u'_z = \frac{z}{t} = \frac{u_z \sqrt{1 - v^2/c^2}}{1 - vu_x/c^2}.$$

Four-vectors

It is convenient at this point to introduce some simplifying notation. We define the *position-time four-vector* x^μ , $\mu = 0, 1, 2, 3$, as follows:

$$x^0 = ct, \quad x^1 = x, \quad x^2 = y, \quad x^3 = z$$

In terms of x^μ , the Lorentz transformations take on a more symmetrical appearance:

$$x^{0'} = \gamma(x^0 - \beta x^1)$$

$$x^{1'} = \gamma(x^1 - \beta x^0)$$

$$x^{2'} = x^2$$

$$x^{3'} = x^3$$

where

$$\beta \equiv \frac{v}{c}$$

More compactly:

$$x^{\mu'} = \sum_{\nu=0}^3 \Lambda_{\nu}^{\mu'} x^{\nu} \quad (\mu = 0, 1, 2, 3)$$

The coefficients Λ_{ν}^{μ} may be regarded as the elements of a matrix Λ :

$$\Lambda = \begin{bmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(i.e. $\Lambda_0^0 = \Lambda_1^1 = \gamma$; $\Lambda_0^1 = \Lambda_1^0 = -\gamma\beta$; $\Lambda_2^2 = \Lambda_3^3 = 1$; and all the rest are zero). To avoid writing lots of Σ 's, we shall follow Einstein's 'summation convention', which says that repeated Greek indices (one as subscript, one as superscript) are to be summed from 0 to 3. Thus

$$x^{\mu'} = \Lambda_{\nu}^{\mu'} x^{\nu}$$

A special virtue of this tidy notation is that the same form describes Lorentz transformations that are *not* along the x direction; in fact, the S and S' axes need not even be parallel; the Λ matrix is more complicated

Although the individual coordinates of an event change, when we go from S to S' there is a particular *combination* of them that remains the same

$$I \equiv (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2 = (x^{0'})^2 - (x^{1'})^2 - (x^{2'})^2 - (x^{3'})^2$$

Four-vectors

$$a \cdot b = a^0 b^0 - \mathbf{a} \cdot \mathbf{b}$$

Such a quantity, which has the same value in any inertial system, is called an *invariant*. (In the same sense, the quantity $r^2 = x^2 + y^2 + z^2$ is invariant under *rotations*.) Now, I would *like* to write this invariant in the form of a sum: $\sum_{\mu=0}^3 x^\mu x^\mu$, but unfortunately there are those three irritating minus signs. To keep track of them, we introduce the *metric*, $g_{\mu\nu}$, whose components can be displayed as a matrix $\mathbf{g}$:

$$\mathbf{g} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

(i.e. $g_{00} = 1$; $g_{11} = g_{22} = g_{33} = -1$; all the rest are zero).^{*} With the help of $g_{\mu\nu}$, the invariant I can be written as a double sum:

$$I = \sum_{\mu=0}^3 \sum_{\nu=0}^3 g_{\mu\nu} x^\mu x^\nu = g_{\mu\nu} x^\mu x^\nu$$

Carrying things a step further, we define the *covariant* four-vector x_μ (index *down*) as follows:

$$x_\mu \equiv g_{\mu\nu} x^\nu$$

(i.e. $x_0 = x^0$, $x_1 = -x^1$, $x_2 = -x^2$, $x_3 = -x^3$). To emphasize the distinction we call the ‘original’ four-vector x^μ (index *up*) a *contravariant* four-vector. The invariant I can then be written in its cleanest form:

$$I = x_\mu x^\mu$$

The position-time four-vector x^μ is the archetype for all *four-vectors*. We define a four-vector, a^μ , as a four-component object that transforms in the same way x^μ does when we go from one inertial system to another, to wit:

$$a^{\mu'} = \Lambda^{\mu}_{\nu} a^\nu$$

with the same coefficients Λ^{μ}_{ν} . To each such (contravariant) four-vector we associate a *covariant* four-vector a_μ , obtained by simply changing the signs of the spatial components, or, more formally

$$a_\mu = g_{\mu\nu} a^\nu$$

Of course, we can go back from covariant to contravariant by reversing the signs again:

$$a^\mu = g^{\mu\nu} a_\nu$$

where $g^{\mu\nu}$ are technically the elements in the matrix $\mathbf{g}^{-1}$ (however, since our metric is its own inverse, $g^{\mu\nu}$ is the same as $g_{\mu\nu}$). Given any two four-vectors, a^μ and b^μ , the quantity

$$a^\mu b_\mu = a_\mu b^\mu = a^0 b^0 - a^1 b^1 - a^2 b^2 - a^3 b^3$$

is invariant (the same number in any inertial system). We shall refer to it as the *scalar product* of a and b : it is the four-dimensional analog to the dot product of two three-vectors

Four-vectors

We also use the notation a^2 for the scalar product of a^μ with itself:[†]

$$a^2 \equiv a \cdot a = (a^0)^2 - \mathbf{a}^2$$

Notice, however, that a^2 need not be positive. Indeed, we can classify all four-vectors according to the sign of a^2 :

$$\begin{aligned} \text{If } a^2 > 0, & \quad a^\mu \text{ is called } \textit{timelike} \\ \text{If } a^2 < 0, & \quad a^\mu \text{ is called } \textit{spacelike} \\ \text{If } a^2 = 0, & \quad a^\mu \text{ is called } \textit{lightlike} \end{aligned}$$

From *vectors* it is a short step to *tensors*: a second-rank tensor, $s^{\mu\nu}$, carries two indices, has $4^2 = 16$ components, and transforms with *two* factors of Λ :

$$s^{\mu\nu'} = \Lambda_\kappa^\mu \Lambda_\sigma^\nu s^{\kappa\sigma}$$

a third-rank tensor, $t^{\mu\nu\lambda}$, has three indices, $4^3 = 64$ components, and transforms with *three* factors of Λ :

$$t^{\mu\nu\lambda'} = \Lambda_\kappa^\mu \Lambda_\sigma^\nu \Lambda_\tau^\lambda t^{\kappa\sigma\tau}$$

and so on. In this hierarchy, a vector is a tensor of rank one, and a scalar (invariant) is a tensor of rank zero. We construct covariant and ‘mixed’ tensors by lowering indices (at the cost of a minus sign for each spatial index), for example

$$s^\mu{}_\nu = g_{\nu\lambda} s^{\mu\lambda}; \quad s_{\mu\nu} = g_{\mu\kappa} g_{\nu\lambda} s^{\kappa\lambda}$$

Energy and Momentum

Suppose you’re driving down the highway, and pretend for the sake of argument that you’re going at close to the speed of light. You might want to keep an eye on two different ‘times’: if you’re worried about making an appointment in San Francisco, you should check the stationary clocks posted now and then along the side of the road. But if you’re wondering when would be an appropriate time to stop for a bite to eat, it would be more sensible to look at the watch on your wrist. For, according to relativity, the moving clock (in this case, your watch) is running slow (relative to the ‘stationary’ clocks on the ground), and so too is your heart rate, your metabolism, your speech and thought, *everything*. Specifically, while the ‘ground’ time advances by an infinitesimal amount dt , your own (or *proper*) time advances by the smaller amount $d\tau$:

$$d\tau = \frac{dt}{\gamma}$$

At normal driving speeds, of course, γ is so close to 1 that dt and $d\tau$ are essentially identical, but in elementary particle physics the distinction between laboratory time (read off the clock on the wall) and particle time (as it would appear on the particle’s watch) is crucial. Although we can always get from one to the other, using in practice it is usually most convenient to work with proper time, *because τ is invariant* – all observers can read the particle’s watch, and at any given moment they must all agree on what it says, even though their *own* clocks may differ from it and from one another.

Four-vectors

When we speak of the ‘velocity’ of a particle (with respect to the laboratory), we mean, of course, the distance it travels (measured in the lab frame) divided by the time it takes (measured on the lab clock):

$$\mathbf{v} = \frac{d\mathbf{x}}{dt}$$

But in view of what has just been said, it is also useful to introduce the *proper velocity*, $\boldsymbol{\eta}$, which is the distance traveled (again, measured in the lab frame) divided by the *proper* time:*

$$\boldsymbol{\eta} \equiv \frac{d\mathbf{x}}{d\tau}$$

the two velocities are related by a factor of γ :

$$\boldsymbol{\eta} = \gamma \mathbf{v}$$
$$\eta^\mu = \frac{dx^\mu}{d\tau}$$

whose zeroth component is

$$\eta^0 = \frac{dx^0}{d\tau} = \frac{d(ct)}{(1/\gamma) dt} = \gamma c$$

Thus

$$\eta^\mu = \gamma (c, v_x, v_y, v_z)$$

Incidentally, $\eta_\mu \eta^\mu$ should be invariant, and it is:

$$\eta_\mu \eta^\mu = \gamma^2 (c^2 - v_x^2 - v_y^2 - v_z^2) = \gamma^2 c^2 (1 - v^2/c^2) = c^2$$

They don’t make ’em more invariant than *that*!

Classically, momentum is mass times velocity. We would like to carry this over in relativity, but the question arises: which velocity should we use – *ordinary* velocity or *proper* velocity? Classical considerations offer no clue, for the two are equal in the nonrelativistic limit. In a sense, it’s just a matter of definition, but there is a subtle and compelling reason why ordinary velocity would be a *bad* choice, whereas proper velocity is a *good* choice. The point is this: if we defined momentum as $m\mathbf{v}$, then the law of conservation of momentum would be inconsistent with the principle of relativity (if it held in one inertial system, it would *not* hold in other inertial systems). But, if we define momentum as $m\boldsymbol{\eta}$, then conservation of momentum *is* consistent with the principle of relativity (if it holds in one inertial system, it automatically holds in all inertial systems)

$$\mathbf{p} \equiv m\boldsymbol{\eta}$$

Since proper velocity is part of a four-vector, the same goes for momentum:

$$p^\mu = m\eta^\mu$$

The spatial components of p^μ constitute the (relativistic) momentum three-vector:

$$\mathbf{p} = \gamma m\mathbf{v} = \frac{m\mathbf{v}}{\sqrt{1 - v^2/c^2}}$$

Four-vectors

Meanwhile, the ‘temporal’ component is

$$p^0 = \gamma mc$$

For reasons that will appear in a moment, we define the *relativistic energy*, E , as

$$E \equiv \gamma mc^2 = \frac{mc^2}{\sqrt{1 - v^2/c^2}}$$

The zeroth component of p^μ , then, is E/c . Thus, energy and momentum together make up a four-vector – the *energy–momentum four-vector* (or *four-momentum*)

$$p^\mu = \left(\frac{E}{c}, p_x, p_y, p_z \right)$$

Incidentally,

$$p_\mu p^\mu = \frac{E^2}{c^2} - \mathbf{p}^2 = m^2 c^2$$

which, again, is manifestly invariant.

The relativistic momentum reduces to the classical expression in the nonrelativistic regime ($v \ll c$), but the same cannot be said for relativistic energy. To see how this quantity comes to be called ‘energy,’ we expand the radical in a Taylor series:

$$E = mc^2 \left(1 + \frac{1}{2} \frac{v^2}{c^2} + \frac{3}{8} \frac{v^4}{c^4} + \dots \right) = mc^2 + \frac{1}{2} mv^2 + \frac{3}{8} m \frac{v^4}{c^2} + \dots$$

Given two four-vectors, $a^\mu = (3, 4, 1, 2)$ and $b^\mu = (5, 0, 3, 4)$, find: a_μ , b_μ , a^2 , b^2 , $a \cdot b$, a^2 , b^2 , and $a \cdot b$. Characterize a^μ and b^μ as timelike, spacelike, or lightlike.

Is p^μ timelike, spacelike, or lightlike for a (real) particle of mass m ? How about a massless particle? How about a virtual particle?

Note that the space-like components of A^μ and A_μ are $\mathbf{A}$ and $-\mathbf{A}$, respectively. The exception is

$$\partial^\mu = \left(\frac{\partial}{\partial t}, -\nabla \right) \quad \text{and} \quad \partial_\mu = \left(\frac{\partial}{\partial t}, \nabla \right),$$

which can be shown to transform like $x^\mu = (t, \mathbf{x})$ and $x_\mu = (t, -\mathbf{x})$, respectively. Thus, the covariant form

$$p^\mu \rightarrow i\partial^\mu.$$

From ∂_μ and ∂^μ we can form the invariant (D’Alembertian) operator

$$\square^2 \equiv \partial_\mu \partial^\mu.$$

Nonrelativistic Quantum Mechanics

We begin by recalling that a prescription for obtaining the Schrödinger equation for a free particle of mass m is to substitute into the classical energy momentum relation

$$E = \frac{p^2}{2m}$$

the differential operators

$$E \rightarrow i\hbar \frac{\partial}{\partial t}, \quad \mathbf{p} \rightarrow -i\hbar \nabla.$$

The resulting operator equation is understood to act on a (complex) wavefunction $\psi(\mathbf{x}, t)$. That is (with $\hbar \equiv 1$),

$$i \frac{\partial \psi}{\partial t} + \frac{1}{2m} \nabla^2 \psi = 0,$$

where we interpret

$$\rho = |\psi|^2$$

as the probability density ($|\psi|^2 d^3x$ gives the probability of finding the particle in a volume element d^3x).

We are often concerned with moving particles, for example, the collision of one particle with another. We therefore need to be able to calculate the density flux of a beam of particles, $\mathbf{j}$. Now from the conservation of probability, the rate of decrease of the number of particles in a given volume is equal to the total flux of particles out of that volume, that is,

$$-\frac{\partial}{\partial t} \int_V \rho dV = \int_S \mathbf{j} \cdot \hat{\mathbf{n}} dS = \int_V \nabla \cdot \mathbf{j} dV$$

where the last equality is Gauss's theorem and $\hat{\mathbf{n}}$ is a unit vector along the outward normal to the element dS of the surface S enclosing volume V . The probability and the flux densities are therefore related by the “continuity” equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0.$$

To determine the flux, we first form $\partial \rho / \partial t$ by subtracting the wave equation, multiplied by $-i\psi^*$ from the complex conjugate equation multiplied by $-i\psi$. We then obtain

$$\frac{\partial \rho}{\partial t} - \frac{i}{2m} (\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*) = 0.$$

we identify the probability flux density as

$$\mathbf{j} = -\frac{i}{2m} (\psi^* \nabla \psi - \psi \nabla \psi^*).$$

For example, a solution

$$\psi = N e^{i\mathbf{p} \cdot \mathbf{x} - iEt},$$

which describes a free particle of energy E and momentum $\mathbf{p}$, has

$$\rho = |N|^2, \quad \mathbf{j} = \frac{\mathbf{p}}{m} |N|^2.$$

Klein-Gordon Equation

$$E^2 = \mathbf{p}^2 + m^2.$$

Making the operator substitutions, we obtain

$$-\frac{\partial^2 \phi}{\partial t^2} + \nabla^2 \phi = m^2 \phi,$$

which is known as the Klein-Gordon equation (but could, more correctly, have been called the relativistic Schrödinger equation). Multiplying the Klein-Gordon equation by $-i\phi^*$ and the complex conjugate equation by $-i\phi$, and subtracting, gives the relativistic analogue

$$\underbrace{\frac{\partial}{\partial t} \left[i \left(\phi^* \frac{\partial \phi}{\partial t} - \phi \frac{\partial \phi^*}{\partial t} \right) \right]}_{\rho} + \nabla \cdot \underbrace{[-i(\phi^* \nabla \phi - \phi \nabla \phi^*)]}_{\mathbf{j}} = 0.$$

By comparison we identify the probability and the flux densities with the terms in square brackets. For example, for a free particle of energy E and momentum $\mathbf{p}$, described by the Klein-Gordon solution

$$\phi = Ne^{i\mathbf{p} \cdot \mathbf{x} - iEt},$$

we find

$$\rho = i(-2iE)|N|^2 = 2E|N|^2$$

$$\mathbf{j} = -i(2i\mathbf{p})|N|^2 = 2\mathbf{p}|N|^2.$$

We see that the probability density is proportional to E , the relativistic energy of the particle. (We defer the explanation of this for a moment.)

It is advantageous to express these results in four-vector notation. Not only are they then more concise, but also the covariance becomes explicit. Using the D'Alembertian operator, the Klein-Gordon equation becomes

$$(\square^2 + m^2)\phi = 0.$$

Moreover, the probability and the flux densities form a four-vector

$$j^\mu = (\rho, \mathbf{j}) = i(\phi^* \partial^\mu \phi - \phi \partial^\mu \phi^*)$$

which satisfies the (covariant) continuity relation

$$\partial_\mu j^\mu = 0.$$

Taking the free particle solution

$$\phi = Ne^{-ip \cdot x},$$

we have

$$j^\mu = 2p^\mu |N|^2.$$

We noted that the probability density ρ is the time-like component of a four-vector; ρ is proportional to E . This result may be anticipated since under a Lorentz boost of velocity $\mathbf{v}$, a volume element suffers a Lorentz contraction $d^3x \rightarrow d^3x\sqrt{1-v^2}$; and so, to keep ρd^3x invariant, we require ρ to transform as the time-like component of a four-vector $\rho \rightarrow \rho/\sqrt{1-v^2}$.

So far, so good; but what are the energy eigenvalues of the Klein-Gordon equation?

$$E = \pm(\mathbf{p}^2 + m^2)^{1/2}.$$

Thus, in addition to the acceptable $E > 0$ solutions, we have negative energy solutions. This looks at first like a total disaster, because transitions can occur to lower and lower (more negative) energies. A second problem is that the $E < 0$ solutions are associated with a negative probability density. To summarize, the difficulties are

$E < 0 \text{ solutions with } \rho < 0.$

It is clear that this problem cannot be simply ignored. We cannot simply discard the negative energy solutions as we have to work with a complete set of states, and this set inevitably includes the unwanted states.

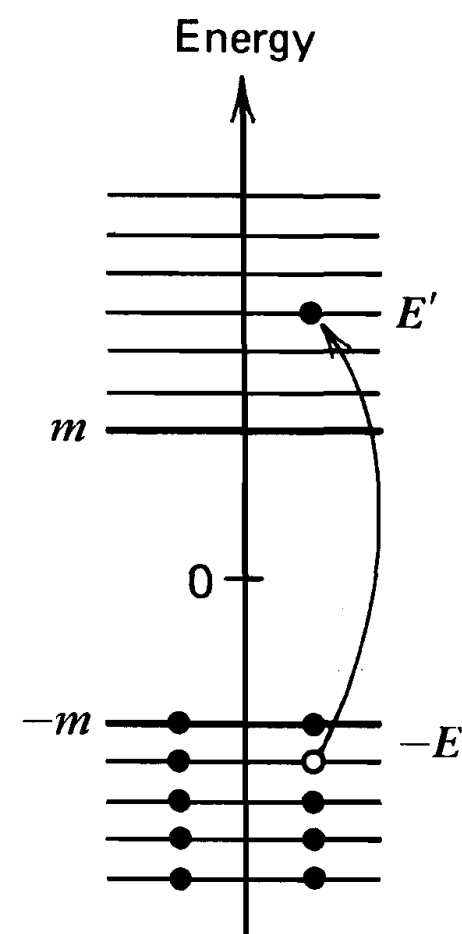
Dirac's Idea

In 1927, in an attempt to avoid these problems, Dirac devised a relativistic wave equation linear in $\partial/\partial t$ and ∇ . He succeeded in overcoming the problem of the negative probability density, with the unexpected bonus that the equation described spin- $\frac{1}{2}$ particles. However, $E < 0$ solutions still occurred, as can be seen in the energy spectrum for a free Dirac electron

Dirac sidestepped the negative energy solutions by invoking the exclusion principle. He postulated that all the negative energy states are occupied and regarded the vacuum as an infinite sea of $E < 0$ electrons. Now the positive energy electrons cannot collapse into the lower (negative) energy levels, as this is prevented by the exclusion principle. One can, however, create a “hole” in the sea by excitation of an electron from a negative energy ($-E$) state to a positive energy (E') state, as shown. The absence of an electron of charge $-e$ and energy $-E$ is interpreted as the presence of an antiparticle (a positron) of charge $+e$ and energy $+E$. Thus, the net effect of this excitation is the production of a pair of particles

$$e^-(E') + e^+(E),$$

which clearly requires energy $E + E' \geq 2m$ (see diagram). Until 1934, the Dirac equation was considered to be the only acceptable relativistic wave equation.



In 1934, Pauli and Weisskopf revived the Klein–Gordon equation by inserting the charge $-e$ into j^μ and interpreting it as the charge-current density of the electron,

$$j^\mu = -ie(\phi^* \partial^\mu \phi - \phi \partial^\mu \phi^*).$$

Now, $\rho = j^0$ represents a charge density, not a probability density, and so the fact that it can be negative is no longer objectionable. In some sense, which we shall make clear in a moment, the $E < 0$ solutions may then be regarded as $E > 0$ solutions for particles of opposite charge (antiparticles). Unlike “hole theory,” this interpretation is applicable to bosons as well as fermions. One cannot fill up the Dirac sea with bosons, as there is no exclusion principle operative to stack the particles. To develop the antiparticle idea and to introduce Feynman diagrams, it is useful to first ignore the complications due to the spin of the electrons.

The Feynman–Stückelberg Interpretation of $E < 0$ Solutions

The prescription we use for handling negative energy states was proposed by Stückelberg (1941) and by Feynman (1948). Expressed most simply, the idea is that a negative energy solution describes a particle which propagates backward in time or, equivalently, a positive energy antiparticle propagating forward in time.

Feynman-Stückelberg Interpretation

Consider an electron of energy E , three-momentum $\mathbf{p}$, and charge $-e$. we know that the electromagnetic four-vector current is

$$j^\mu(e^-) = -2e|N|^2(E, \mathbf{p}).$$

Now take an antiparticle, a positron, with the same $E, \mathbf{p}$. Since its charge is $+e$,

$$\begin{aligned} j^\mu(e^+) &= +2e|N|^2(E, \mathbf{p}) \\ &= -2e|N|^2(-E, -\mathbf{p}), \end{aligned}$$

which is exactly the same as the current j^μ for an electron with $-E, -\mathbf{p}$. Thus, as far as a system is concerned, the emission of a positron with energy E is the same as the absorption of an electron of energy $-E$. Pictorially, we have

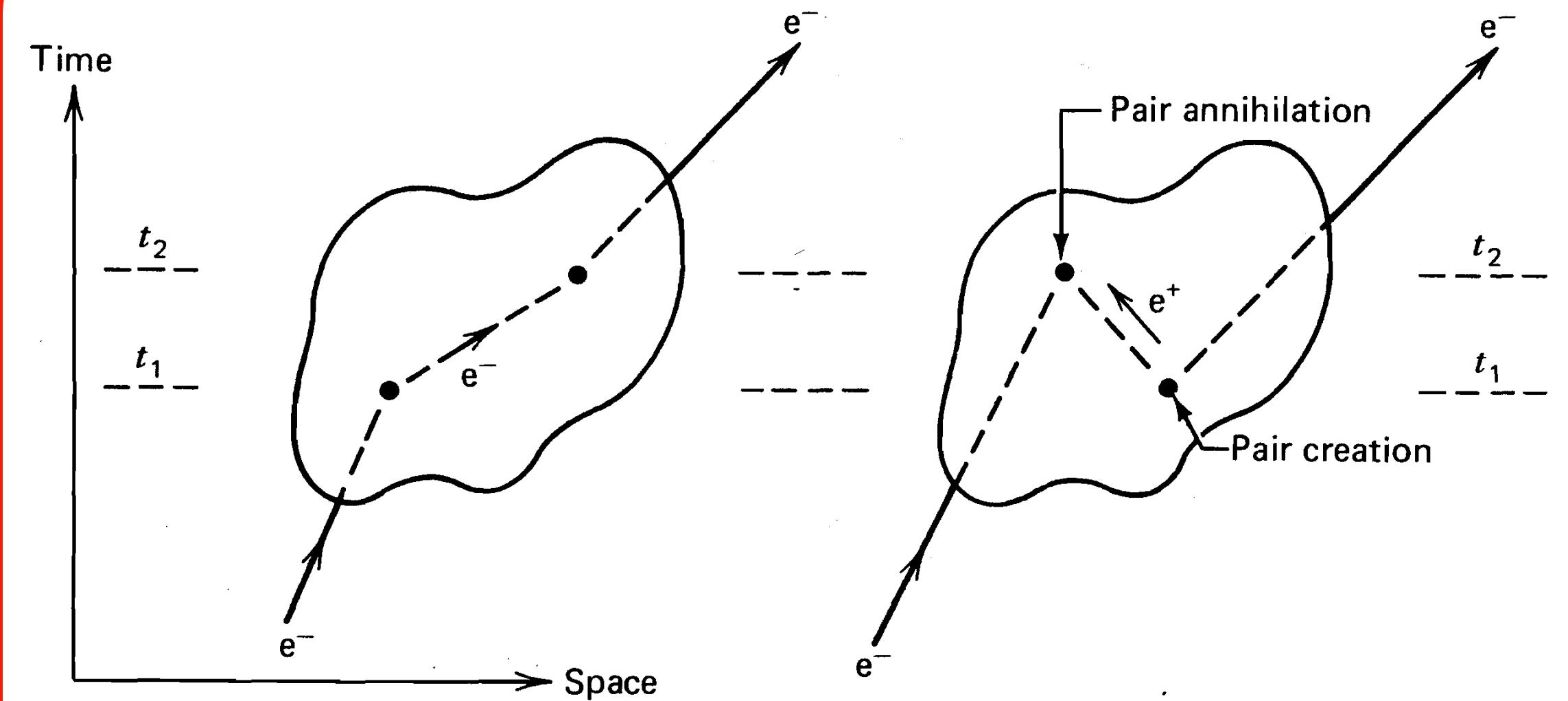
$$\boxed{\begin{array}{c} \uparrow e^+ \\ E > 0 \end{array} \equiv \begin{array}{c} \downarrow e^- \\ (-E) < 0 \end{array}} \quad \uparrow \text{time}$$

In other words, negative-energy *particle* solutions going backward in time describe positive-energy *antiparticle* solutions going forward in time. Of course, the reason why this identification can be made is simply because

$$e^{-i(-E)(-t)} = e^{-iEt}.$$

The single-particle (e^-) wavefunction formalism not only handles antiparticles but can even describe many-particle situations. As an example, we consider the double scattering of an electron in a potential.

The crucial observation is that there are two pictures corresponding to the same observation. There are two different time orderings of the two interactions with the potential that lead to the same observable event. Indeed, note that the (observable) path of the electron before and after the double scattering is the same in the two diagrams. The second



picture is only possible because of the antiparticle prescription. At time t_2 , the electron scatters backward in time (with $E < 0$). This electron is interpreted as a positron (with $E > 0$) going forward in time. Then, the events in the diagram can be viewed as follows: first, at time t_1 , an e^-e^+ pair is created; then, at a later time t_2 , the e^+ is annihilated by the incident e^- . Therefore, between t_1 and t_2 , the electron trajectory drawn in the second diagram actually describes three particles: the initial and final electrons and a positron! As both double scatterings lead to the same observed final electron, they both have to be included in computing the probability of this event. Note that, just as in hole theory, the vacuum has become a very complex environment: e^-e^+ pairs can pop out of it and disappear into it as a result of the antiparticle prescription!

Perturbation Theory

Suppose we know the solutions to the free-particle Schrödinger equation

$$H_0 \phi_n = E_n \phi_n \quad \text{with} \quad \int_V \phi_m^* \phi_n d^3x = \delta_{mn}$$

where H_0 , the Hamiltonian, is time independent. For simplicity, we have normalized the solutions to one particle in a box of volume V . The objective is to solve Schrödinger's equation

$$(H_0 + V(\mathbf{x}, t))\psi = i \frac{\partial \psi}{\partial t}$$

for a particle moving in the presence of an interaction potential $V(\mathbf{x}, t)$.

Any solution can be expressed in the form

$$\psi = \sum_n a_n(t) \phi_n(\mathbf{x}) e^{-iE_n t}.$$

Now, to find the unknown coefficients $a_n(t)$, we obtain

$$i \sum_n \frac{da_n}{dt} \phi_n(\mathbf{x}) e^{-iE_n t} = \sum_n V(\mathbf{x}, t) a_n \phi_n(\mathbf{x}) e^{-iE_n t}.$$

Multiplying by ϕ_f^* , integrating over the volume, and using the orthonormality relation leads to the following coupled linear differential equations for the a_n coefficients:

$$\frac{da_f}{dt} = -i \sum_n a_n(t) \int \phi_f^* V \phi_n d^3x e^{i(E_f - E_n)t}.$$

Suppose that before the potential V acts the particle is in an eigenstate i of the unperturbed Hamiltonian, that is, at time $t = -T/2$:

$$\begin{aligned} a_i(-T/2) &= 1, \\ a_n(-T/2) &= 0 \quad \text{for } n \neq i, \end{aligned}$$

and

$$\frac{da_f}{dt} = -i \int d^3x \phi_f^* V \phi_i e^{i(E_f - E_i)t}.$$

Now, provided that the potential is small and transient, we can, as a first approximation, assume that these initial conditions remain true at all times. Then,

$$a_f(t) = -i \int_{-T/2}^t dt' \int d^3x \phi_f^* V \phi_i e^{i(E_f - E_i)t'}$$

and, in particular, at time $t = +T/2$ after the interaction has ceased,

$$T_{fi} \equiv a_f(T/2) = -i \int_{-T/2}^{T/2} dt \int d^3x [\phi_f(\mathbf{x}) e^{-iE_f t}]^* V(\mathbf{x}, t) [\phi_i(\mathbf{x}) e^{-iE_i t}]$$

which we may write in the covariant form

$$T_{fi} = -i \int d^4x \phi_f^*(x) V(x) \phi_i(x).$$

Of course, the expression for $a_f(t)$ is only a good approximation if $a_f(t) \ll 1$, as this has been assumed in obtaining the result.

We are tempted to interpret $|T_{fi}|^2$ as the probability that the particle is scattered from an initial state i to a final state f . Is this interpretation valid? Consider the case when $V(\mathbf{x}, t) = V(\mathbf{x})$ is time independent; then,

$$\begin{aligned} T_{fi} &= -i V_{fi} \int_{-\infty}^{\infty} dt e^{i(E_f - E_i)t} \\ &= -2\pi i V_{fi} \delta(E_f - E_i) \end{aligned}$$

with

$$V_{fi} \equiv \int d^3x \phi_f^*(\mathbf{x}) V(\mathbf{x}) \phi_i(\mathbf{x}).$$

Perturbation Theory

The δ -function expresses the fact that the energy of the particle is conserved in the transition $i \rightarrow f$. By the uncertainty principle, this means that an infinite time separates the states i and f , and $|T_{fi}|^2$ is therefore not a meaningful quantity. We define instead a transition probability per unit time

$$W = \lim_{T \rightarrow \infty} \frac{|T_{fi}|^2}{T}.$$

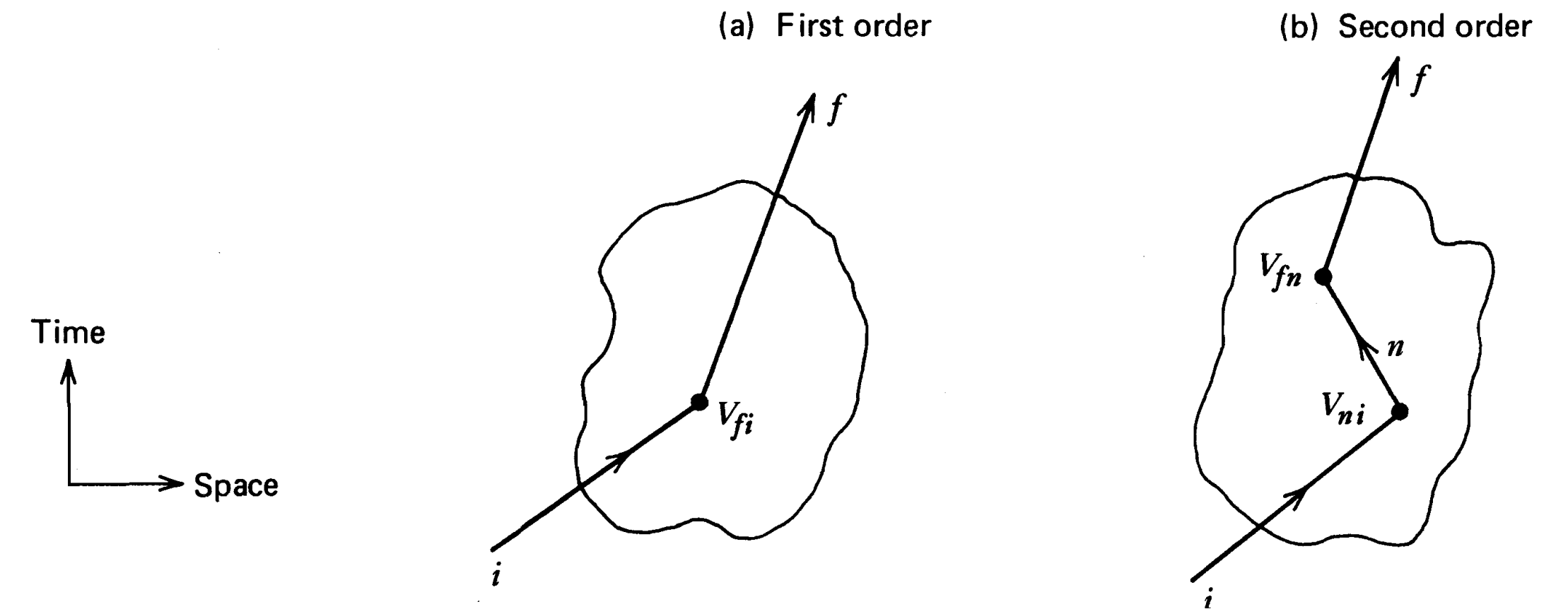
Squaring

$$\begin{aligned} W &= \lim_{T \rightarrow \infty} 2\pi \frac{|V_{fi}|^2}{T} \delta(E_f - E_i) \int_{-T/2}^{+T/2} dt e^{i(E_f - E_i)t} \\ &= \lim_{T \rightarrow \infty} 2\pi \frac{|V_{fi}|^2}{T} \delta(E_f - E_i) \int_{-T/2}^{+T/2} dt \\ &= 2\pi |V_{fi}|^2 \delta(E_f - E_i). \end{aligned}$$

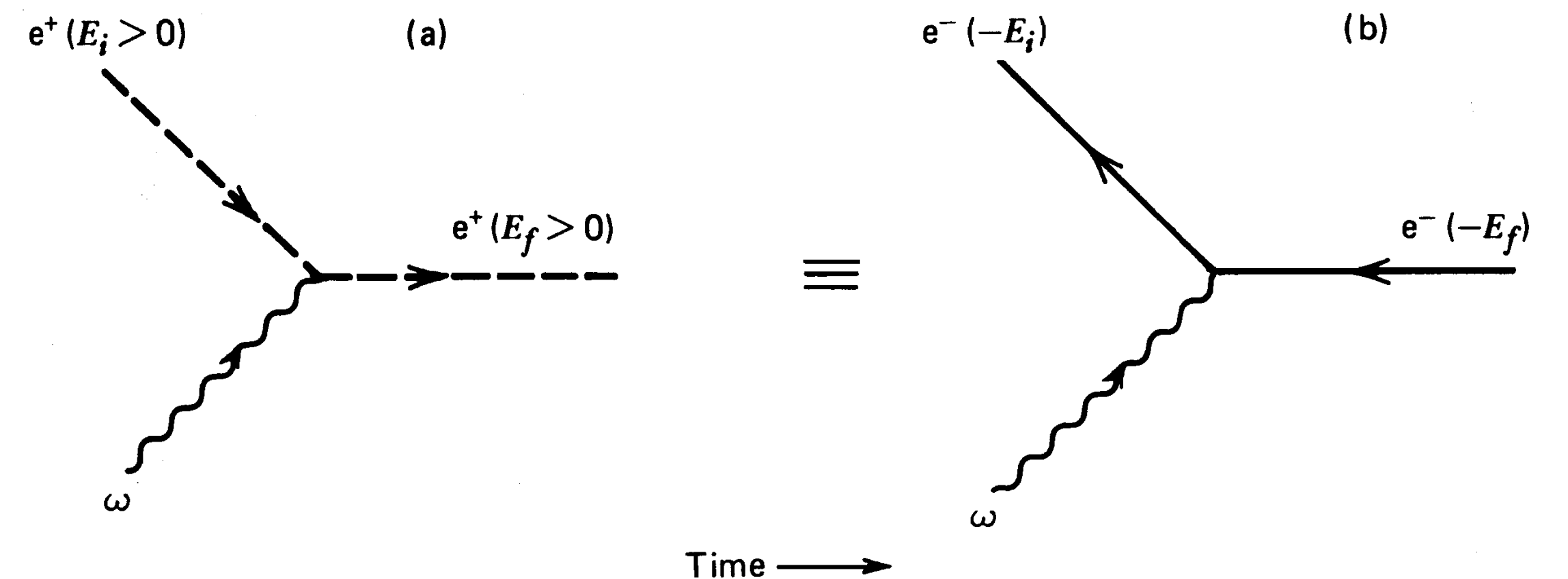
This equation can only be given physical meaning after integrating over a set of initial and final states. In particle physics, we usually deal with situations where we start with a specified initial state and end up in one of a set of final states. Let $\rho(E_f)$ be the density of final states; that is, $\rho(E_f) dE_f$ is the number of states in the energy interval E_f to $E_f + dE_f$. We integrate over this density, imposing energy conservation, and obtain the transition rate

$$\begin{aligned} W_{fi} &= 2\pi \int dE_f \rho(E_f) |V_{fi}|^2 \delta(E_f - E_i) \\ &= 2\pi |V_{fi}|^2 \rho(E_i). \end{aligned}$$

This is Fermi's Golden Rule.



$$\int (e^{-iE_f t})^* e^{-i\omega t} e^{-iE_i t} dt = 2\pi \delta(E_f - \omega - E_i) \quad \int (e^{-i(-E_i)t})^* e^{-i\omega t} e^{-i(-E_f)t} dt = 2\pi \delta(-E_i - \omega + E_f)$$



The *rule* is to form the matrix element

$$\int \phi_{\text{outgoing}}^* V \phi_{\text{ingoing}} d^4x.$$